

Name:

Deep Learning



STN:

Computer Assignment 2

Persian Handwritten Digit Recognition Using Fully Connected Neural Networks

The objective of this assignment is to design a model using fully connected neural networks. In this exercise, we use the HODA Persian handwritten digit dataset. The dataset provided to you contains 10 classes of handwritten digits (0-9). Use 10,000 images for training and 2,000 images for testing.

You must design a network that can recognize which of these classes an input image belongs to. To do this, we will design a fully connected network. First, choose the network architecture as you wish and try to obtain the best possible accuracy by modifying it. Report your results.

- 1) First, resize the available images to 20x20 pixels and perform the necessary preprocessing on them before feeding them into the neural network.
- 2) Choose appropriate hyperparameters for the network and explain how you determined these hyperparameters.
- 3) Train the network and report its accuracy on the training data.
- 4) Plot the error curve during the training process.
- 5) After training, evaluate the proposed model on the test data using evaluate.
- 6) Using the trained model and the predict function, make predictions on the test dataset. Display some of the images for which the model made incorrect predictions. In your opinion, what is the cause of the model's error on these images?
- 7) For a better understanding of the prediction function, apply it to the first test image and examine the probabilities shown. Did the model predict the image correctly? Show the true identity (digit) of the number.

Now that you have designed and trained your fully connected model, answer the following questions:

- 1) What is the highest accuracy you can achieve using the fully connected model?
- 2) What are the best optimizer and learning rate for the network? Provide results for several different optimizers and learning rates, and compare them with one another.

Additional Notes:

- 1) A starter code has been provided. If you have difficulty running it on your system, use the Google Colab environment.

- 2) There is no need to submit a separate PDF file for the explanations and report; write the Persian explanations in the same .ipynb file.
- 3) Use deep learning libraries such as TensorFlow and Keras to complete this assignment.
- 4) To increase execution speed, you may select a GPU as the processor type.
- 5) This is an individual assignment and copying is not permitted. However, you may consult others about their experience in writing the code and performing analysis and comparison.

Good luck!